



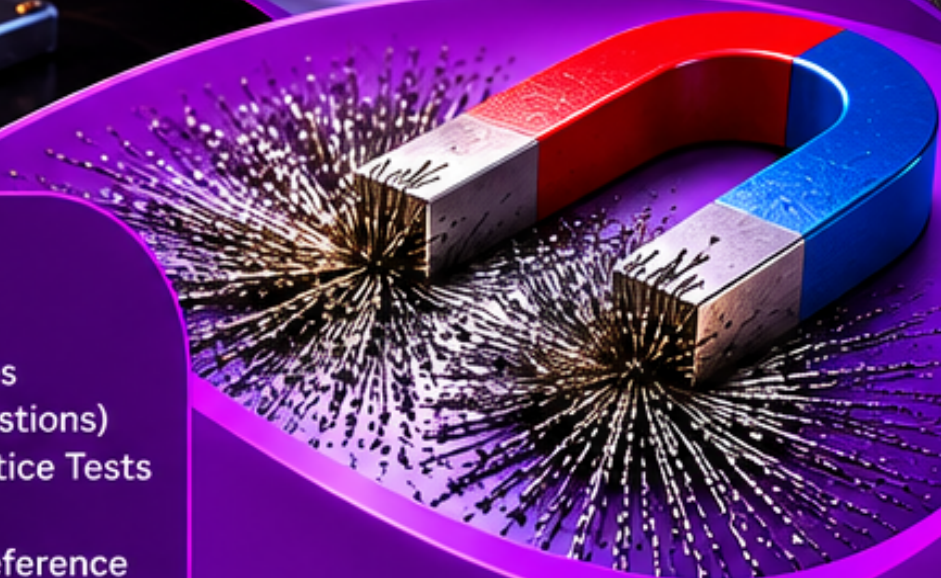
PEAK ACHIEVERS
EDUCATION FOR A BRIGHTER TOMORROW

**COMPETENCY
BASED
CURRICULUM
(CBC)**

PHYSICS

Learner's Book **1**

Grade 10



- ✓ Comprehensive Notes
- ✓ Illustrations and Diagrams
- ✓ Worked Examples
- ✓ Competency-Based Activities
- ✓ Final Mixed Revision (67 Questions)
- ✓ 5 National Examination Practice Tests
- ✓ Marking Schemes
- ✓ Rapid Revision & Formula Reference

Preface

Physics becomes easier when learners can observe, understand, calculate and apply. This revision book has been developed for Grade 10 Senior School Physics within Kenya's Competency-Based Education framework.

The Peak Achievers approach keeps notes brief and learner-friendly, then strengthens understanding through worked examples, practical activities, applications, diagrams, quick checks and revision questions.

Throughout the book, look out for REMEMBER, NB, HINT, QUICK CHECK, APPLICATION, PRACTICAL and THINK BIG features.

Each topic ends with practice. The book then provides final mixed revision, five national examination practice tests, answers and marking guidance, and a rapid formula reference.

UNDERSTAND → PRACTISE → APPLY → ACHIEVE

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How to Use This Book

- 1. Learn:** Read the brief notes and identify the main idea.
 - 2. Remember:** Use REMEMBER, NB and HINT boxes to lock in key points.
 - 3. Follow Examples:** Use Formula → Substitution → Calculation → Answer.
 - 4. Apply:** Connect Physics to real-life situations.
 - 5. Practise:** Attempt Quick Checks, Mini Quizzes and end-of-topic questions.
 - 6. Investigate:** Use practical activities to connect observations with explanations.
 - 7. Think Big:** Attempt reasoning and unfamiliar application problems.
 - 8. Test Yourself:** Complete the mixed revision and five full practice tests.
- PRACTICAL SIGNATURE:** Observation → Interpretation → Conclusion

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Contents

Strand 1 - Mechanics and Thermal Physics

1. Introduction to Physics
2. Pressure
3. Mechanical Properties of Materials
4. Temperature and Thermal Expansion
5. Moments and Equilibrium
6. Energy, Work, Power and Machines

Strand 2 - Waves and Optics

7. Properties of Waves
8. Radioactivity and Stability of Isotopes

Strand 3 - Electricity and Magnetism

9. Electrostatics
10. Current Electricity
11. Introduction to Electronics

Strand 4 - Environmental and Space Physics

12. Greenhouse Effect and Climate Change
13. Introduction to Space Physics

Final Mixed Revision

National Examination Practice Tests 1-5

Answers & Marking Guidance

Rapid Revision & Formula Reference

About the Author

About Peak Achievers Online Academy

Final Learner Message

Page numbers will be generated after final pagination.

CHAPTER 1 - INTRODUCTION TO PHYSICS

What You Need to Know

Physics studies matter, energy and their interactions. It helps us explain events around us and develop useful technologies.

REMEMBER: Physics is not only formulas. It connects observation, measurement, explanation and application.

Major Areas of Physics

- Mechanics
- Thermal physics
- Waves and optics
- Electricity and magnetism
- Electronics
- Space physics

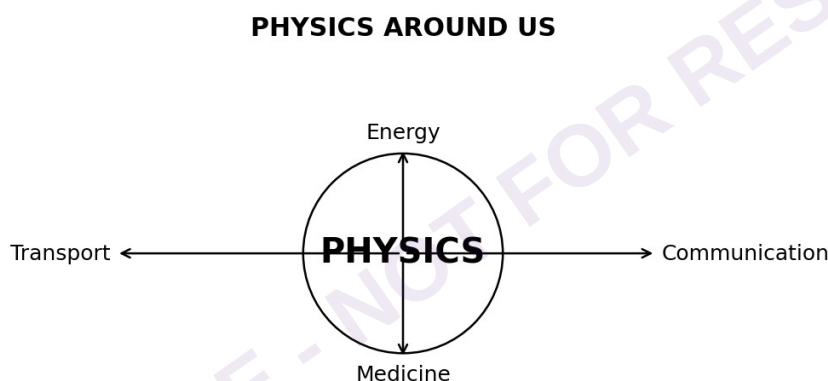


FIGURE 1.1 - PHYSICS AROUND US

Physics and Other Subjects

Physics works closely with Mathematics, Chemistry, Biology and technology. Mathematics helps describe physical relationships and analyse measurements.

Scientific Investigation

Problem → Investigation → Measurement → Analysis → Conclusion

SCIENTIFIC INVESTIGATION FLOW

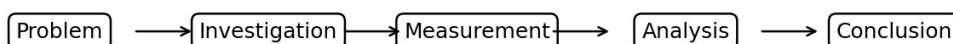


FIGURE 1.2 - SCIENTIFIC INVESTIGATION FLOW

APPLICATION: Physics is used in transport, communication, construction, medicine, energy and space technology.

QUICK CHECK: 1. What is Physics? 2. Name two areas of Physics. 3. Give one application of Physics.

THINK BIG: A learner notices that a shadow changes length when a lamp is moved. Which steps of scientific investigation could be used to study this?

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CHAPTER 2 - PRESSURE

2.1 MEANING AND UNITS OF PRESSURE

- Pressure is the force acting per unit area.
- It depends on the size of the force and the area over which the force acts.

$$P = F/A$$

where P = pressure (Pa), F = force (N), A = area (m²).

SI unit of pressure: pascal (Pa). 1 Pa = 1 N/m².

REMEMBER: Same force, smaller area → greater pressure. Same force, larger area → lower pressure. Pressure depends directly on force and inversely on area.

Worked Example

A force of 600 N acts normally on an area of 0.20 m². Calculate the pressure.

Formula: $P = F/A$

Substitution: $P = 600/0.20$

Calculation: $P = 3000 \text{ Pa}$

Answer: $3.0 \times 10^3 \text{ Pa}$

APPLICATION: Wide tyres on tractors reduce pressure on soft ground. Sharp knives exert high pressure to cut easily. Snowshoes prevent a person from sinking into snow. High-heeled shoes exert greater pressure than flat shoes.

QUICK CHECK 2.1: 1. A force of 600 N acts on an area of 0.20 m². Find the pressure. 2. A force of 100 N acts on a 25 cm² area. Calculate the pressure. 3. State one situation where low pressure is an advantage and one where high pressure is an advantage.

THINK BIG: Why do certain animals, such as camels, have wide feet? Explain in terms of pressure.

2.2 PRESSURE IN A LIQUID

- Liquids exert pressure due to the weight of the liquid above a point.
- Liquid pressure increases with depth.

$$P = \rho gh$$

where P = pressure (Pa), ρ = density of liquid (kg/m³), g = gravitational field strength (m/s²), h = depth (m).

REMEMBER: Liquid pressure increases as depth increases. At the same depth, pressure is the same at all points. Liquid pressure acts in all directions. Liquids exert pressure on the bottom and sides of their containers.

Worked Example

Calculate the pressure at a depth of 5.0 m in water. Take $\rho = 1000 \text{ kg/m}^3$ and $g = 10 \text{ m/s}^2$.

Formula: $P = \rho gh$

Substitution: $P = 1000 \times 10 \times 5.0$

Calculation: $P = 50\,000\text{ Pa}$

Answer: $5.0 \times 10^4\text{ Pa}$

APPLICATION: Pressure in liquids is used in dams, water supply systems and hydraulic machines. Divers experience greater pressure at deeper depths. Submarines are designed to withstand high water pressure.

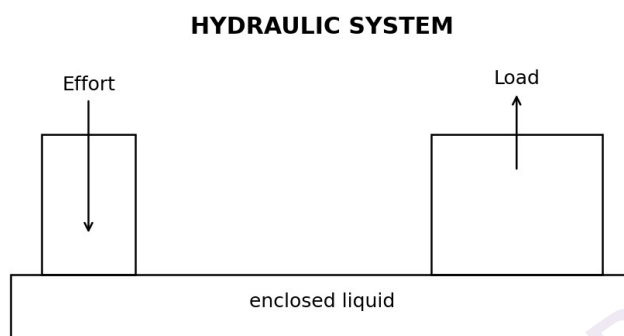


FIGURE 2.1 - SIMPLE HYDRAULIC SYSTEM

QUICK CHECK 2.2: 1. Calculate the pressure at a depth of 5.0 m in water ($\rho = 1000\text{ kg/m}^3$, $g = 10\text{ m/s}^2$). 2. Why do divers need special equipment when going to great depths?

2.3 LIQUID PRESSURE DEMONSTRATION

- The further down a hole, the stronger and further-reaching the jet.
- This shows that liquid pressure increases with depth.

PRACTICAL: Aim: To investigate how liquid pressure changes with depth. Apparatus: Plastic bottle, water and a container. Procedure: Make three similar holes at different heights, cover them, fill the bottle with water, then uncover the holes together. Observation: The lowest jet travels furthest. Interpretation: Pressure is greatest at the greatest depth. Conclusion: Liquid pressure increases with depth.

QUICK CHECK 2.3: 1. Explain why the jet from the lowest hole travels furthest. 2. State one factor that should be kept constant. 3. Suggest one source of error and how to reduce it.

2.4 ATMOSPHERIC PRESSURE

- Air has mass and exerts pressure on surfaces.
- Atmospheric pressure acts in all directions.
- Atmospheric pressure generally decreases as altitude increases.

NB: Many everyday devices work because of a pressure difference between the atmosphere and another region.

APPLICATION: Drinking through a straw, syringes, suction devices and some pumping systems involve pressure differences.

CHAPTER 7 - PROPERTIES OF WAVES

7.1 Meaning of a Wave

A wave is a disturbance that transfers energy from one place to another without a net transfer of matter.

Important wave terms include amplitude, wavelength and frequency.

- Amplitude — maximum displacement from the undisturbed position.
- Wavelength — distance between two successive points in the same phase.
- Frequency — number of complete oscillations or waves produced per second.

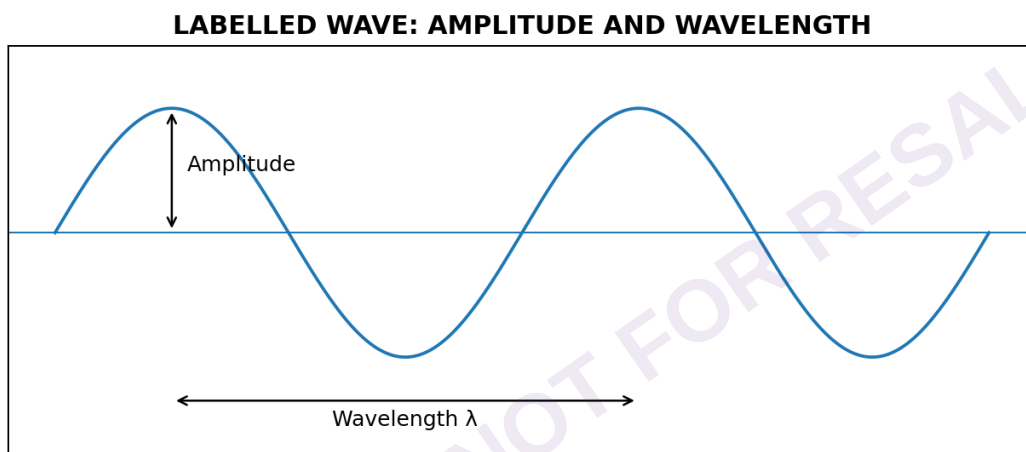


FIGURE 7.1 - LABELLED WAVE: AMPLITUDE AND WAVELENGTH

REMEMBER: Waves transfer energy; the particles of the medium do not travel with the wave from source to destination.

7.2 Rectilinear Propagation

In a uniform medium, waves such as light may travel along straight paths. This behaviour is called rectilinear propagation.

APPLICATION: Straight-line propagation helps explain shadows and the use of line-of-sight optical systems.

7.3 Reflection

Reflection occurs when a wave meets a boundary and returns into the original medium.

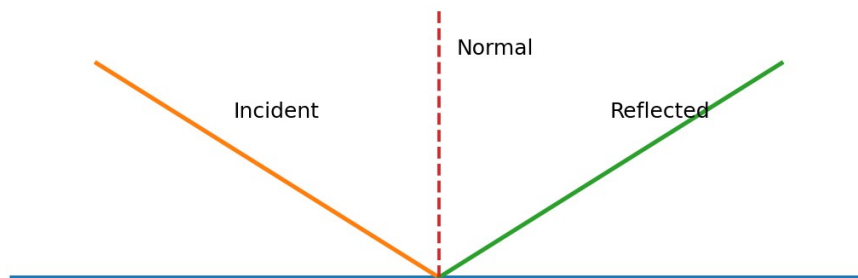
REFLECTION OF A WAVE

FIGURE 7.2 - REFLECTION OF A WAVE

QUICK CHECK: 1. Define a wave. 2. What is amplitude? 3. What happens during reflection?

7.4 Refraction

Refraction is the change in direction of a wave when it passes from one medium or region to another because its speed changes.

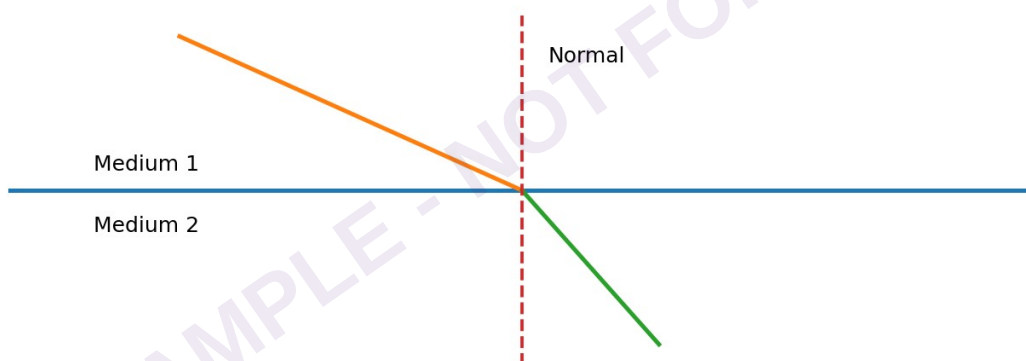
REFRACTION

FIGURE 7.3 - REFRACTION

APPLICATION: Refraction is important in lenses, optical instruments and the apparent displacement of objects viewed through transparent materials.

7.5 Diffraction

Diffraction is the spreading of waves when they pass through an opening or around an obstacle.

The effect is more noticeable when the opening or obstacle is comparable with the wavelength.

21. Two resistors of $6\ \Omega$ and $3\ \Omega$ are connected in parallel across 6 V. Determine their effective resistance and total current. (7 marks)

SECTION C - 30 MARKS

22. Describe an experiment to investigate the V-I relationship of a resistor. Include a labelled circuit, variables, procedure, expected graph and one precaution. (10 marks)

23. A community uses a solar-powered pump to raise water to a tank 8 m above the taps. Calculate the water pressure at the taps and explain two energy transformations and one advantage of the system. Take $\rho = 1000\ \text{kg/m}^3$ and $g = 10\ \text{N/kg}$. (10 marks)

24. Explain Earth's greenhouse effect, distinguish weather from climate, and describe two ways satellites help monitor climate or weather. (10 marks)

TOTAL: 80 MARKS

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